ties based on capability. Greater private participation is expected to improve cost competitiveness and scalability, as demonstrated with the Pinaka multi-barrel rocket launcher, which was jointly developed by DRDO and Tata Advanced Systems Ltd (TASL), with its various components made by TASL, Larsen & Toubro, Solar industries and two DPSUs, Yantra India Ltd and Munitions India Ltd. Missiles in their prototype stages can also be fast-tracked into industrial-scale manufacturing for faster delivery to the armed forces.

Speaking at the India Today Conclave South in Kochi on September 29, G. Satheesh Reddy, member of the National Security Advisory Board and former chairman of DRDO, emphasised that under DRDO's revised ToT framework, technology transfer fees and royalties are waived, making state-backed missile research and patent portfolios accessible to domestic manufacturers. "By integrating private companies from the initial design phase through the DcPP model, DRDO enabled private firms to seamlessly absorb advanced technology and emerge as top integrators, leading to private industry manufacturing for key missile programmes of India like Akash, Astra, Nag ATGMs (anti-tank guided missiles) and MRSAM (medium range surface-to-air missile) sub-assemblies," Reddy said.

A DEFENCE ECOSYSTEM

Military observers point out that long-drawn military engagements like the Russia-Ukraine war or West Asia conflict—where missile shortages have hit belligerents—highlight the requirement for high-volume missile production. Delegating production to private entities ensures the Indian military maintains sufficient stockpiles. Experts also feel that opening up missile production allows the DRDO to focus on next-generation research and testing, while private industry, along with the DPSUs, takes over large-scale production, supply-chain management and lifecycle support.

Private players, on the other hand, now have a level playing field. "With the latest policy decision, private players can pose a challenge to the monopoly of defence PSUs," says M.V. Reddy, MD, Astra Microwave Products, which manufactures sub-systems like telemetry, guidance systems and seekers for Indian missiles. "This will foster a competitive environment, driving down costs and improving delivery schedules."

However, there are several challenges the private sector may have to face. It is one thing to be able to manufacture a complex defence system; it is another to be able to do so repeatedly, with low failure rates, while maintaining consistent quality. That would require an integrated ecosystem linking DRDO, private industry, DPSUs, MSMEs and startups with robust supply chains, the armed forces and export markets.

There is another possible difficulty.

LAUNCHED INTO THE BIG LEAGUE

Here are some key conventional missile systems designed by DRDO that are to be manufactured wholly by private players



LRLACM
Tested in June 2026, the Long Range Land Attack Cruise Missile is a subsonic missile designed to strike targets deep within enemy territory
Range: **1,000–1,500 km**
● *Private firm yet to be finalised*

AKASH AND AKASH-NG
Supersonic surface-to-air missiles that can engage aircraft, drones, cruise & ballistic missiles
Range: **45-50 km** for Akash NG (Next Generation)
● *Private firm yet to be finalised*

ASTRA
Beyond-visual-range (BVR) air-to-air missiles, including Astra Mk-I and Mk-II. Integrated with Sukhoi Su-30 MKI, planned pairing with Rafale
Range: **80-110km** (Mk-I); **240 km** (Mk-II)
● *Made currently by Bharat Dynamics Ltd; Request for Proposal already issued to private sector on Transfer of Technology basis*

RUDRAM
Series of supersonic & hypersonic air-to-surface anti-radiation missiles, including Rudram-I/ II/ III
Range: **100-250 km**
● *Adani Defence noted as Development-cum-Production Partner (DcPP)*



"THE SCALE (OF MISSILE PRODUCTION) WILL DEPEND ON DEMAND FROM THE DEFENCE MINISTRY. NO INDUSTRY WILL INVEST IN FACILITIES ONLY TO KEEP THEM IDLE DUE TO BUREAUCRATIC DELAYS"

Dr RAVI GUPTA
Former DRDO scientist